

Question 1 is about electronic nose

Resistive gas sensors, especially those using semiconducting metal oxides (SMOX), are becoming increasingly popular. They can detect tiny amounts of impurities, are small and stable, and are easy to make and use. These features make SMOX sensors stand out among the many types of gas sensors that work on different physical and chemical principles.

The mixed oxide **X** that has the normal spinel crystal structure, $A^{2+}B^{3+}_2O^{2-}_4$, is an example of one of these SMOX materials. **X** can be obtained by decomposition of the crystalline hydrated metal oxalate $ZC_2O_4 \cdot kH_2O$ of metal Z on air. When heated to 140°C, the crystalline hydrate loses 19.7% of its mass. Further heating in air leads to the formation of 2.407 g of black colored **X** at 500°C and 3.8 L of carbon dioxide at 101325 Pa.

1.1 Determine the formula of **X** and the value of **k** in $ZC_2O_4 \cdot kH_2O$.

In the spinel structures oxygen atoms form a face centered cubic (FCC) lattice. In normal spinels the A^{2+} cations occupy part of the tetrahedral, and the B^{3+} cations occupy part of the octahedral lattice sites.

1.2 Calculate the percentage of the tetrahedral sites that is occupied in the case of a normal spinel $A^{2+}B^{3+}_2O^{2-}_4$.

In addition to the normal spinels, the MM'_2O_4 general formula also describes the inverse and the mixed spinel structures. Here M are the tetrahedrally coordinated, M' the octahedrally coordinated cations, but contrary to normal spinels, particular cations are not restricted to a single lattice site. A characteristic of the inverse spinel structures is that the electron configuration of the M' site cations is d^0 , d^5 or d^{10} .

1.3 Select on the answer sheet two of the given formulas, which correspond to the structure of inverse spinel: Fe_3O_4 , $NiFe_2O_4$, Mn_3O_4 , $FeCr_2O_4$.

Question 2 is about hydrazine salts

Hydrazine (N_2H_4) is a weak diprotic base ($K_{b1} = 8.5 \cdot 10^{-7}$; $K_{b2} = 8.9 \cdot 10^{-16}$). It can form various anhydrous salts with sulfuric acid. The second acid dissociation constant of sulfuric acid is $K_{a2} = 1.02 \cdot 10^{-2}$.

2.1. Provide the theoretically conceivable empirical formulas of anhydrous salts.

2.2. Starting from aqueous solutions, two of these salts can be expected to form. Identify these two salts?

Why doesn't the aqueous procedure work for the other salts?

2.3. Calculate the pH of the 0.1 M solutions of these salts?

Question 3 is about nitrogen oxides

The nitrogen oxides stand out among non-metal oxides, as there is at least one binary molecular oxide corresponding to each oxidation state between +1 and +5.

3.1. Give a molecular formula and a Lewis structure (one resonance structure is sufficient) of the simplest oxide for each oxidation state.

For some oxidation states, multiple oxides are possible. For example, when brown gas **A** is cooled down, a colorless liquid of compound **B** is formed, which is a dimer of **A**. For practical calculations, the volume of the liquid is negligible as compared to the volume of the gas.

3.2. Determine **A** and **B**. Give the molecular formula and a Lewis structure of **B**.

Let us consider the chemical equilibrium $2 \text{A(g)} \rightleftharpoons \text{B(g)}$ (reaction 1) and the following data (assume them to be independent of temperature):

	$\Delta_f H^\circ / \text{kJ/mol}$	$S^\circ / \text{J/(mol}\cdot\text{K)}$
B(g)	9.16	304.29
A(g)	33.18	219.85

3.3. Calculate the equilibrium constant of reaction 1 at 25 °C.

1 mol of **A** is put into an empty container, it is closed with a fixed volume of 25 L and thermostated at 25 °C.

3.4. Calculate the equilibrium pressure of the system.

The standard atmospheric boiling point of **B** is at 21.69 °C. $\Delta H_{\text{vap}} = 22.96 \text{ kJ/mol}$, $\Delta S_{\text{vap}} = 77.9 \text{ J/(mol}\cdot\text{K)}$ for the evaporation of **B**. Assume them to be independent of temperature.

3.5. Calculate the enthalpy of formation of **B(l)**.

3.6. Calculate the entropy change of the reaction $2 \text{A (g)} = \text{B (l)}$ (reaction 2).

We can assume that the entropy change of reaction 2 is independent of temperature between −10 and 22 °C.

3.7. Calculate the equilibrium constant of reaction 2 at 0 °C.

1 mol of **A** is put into an empty container, it is closed with a fixed volume of 25 L and thermostated at 0 °C.

3.8. Calculate the equilibrium pressure of the cooled system.